

CE 310 — Week 13 Assignment: Monte Carlo Simulation

CE 310 | Points: 60 pts (3 problems) | Due: Wed Nov 25, 9:00 AM | Submit as .ipynb (Colab notebook)

The in-class exercise ran a Monte Carlo on one bid item's cost and produced a contingency. This assignment applies the same framework to a scenario the exercise did not cover — one infrastructure (Track A) and one building/energy (Track B). **Complete only the track matching your declared major.**

How to Submit

Read this before you submit. Following this submission format exactly is essential — with a class this size, grading depends on it. Submissions that don't comply will be penalized.

Requirement	Detail
Run all cells	Runtime → Run all — wait until every cell shows a checkmark, no spinners
File format	Download as .ipynb — File → Download → Download .ipynb (not PDF, not .py)
Print lines	Do not edit or delete any <code>print(f"ANSWER_... = ...")</code> line
Seed	Use <code>SEED = 99</code> as set in the template — do not change it
Name and NetID	Fill in NAME and NETID in the identity cell and run it — that cell is what matches your work to your student record. A blank NETID takes a 5-point deduction.
File name	CE310_W13_Assignment_<NetID>.ipynb e.g. CE310_W13_Assignment_abc123.ipynb
Submission	Upload the .ipynb file to D2L → Assignments by 9:00 AM, Wed Nov 25

Setup

```
import numpy as np
import matplotlib.pyplot as plt

MAJOR = "CE" # <-- change to "ArcE" if you are an Architectural Engineering student

SEED = 99 # Different seed from the exercise
N = 10_000
rng = np.random.default_rng(SEED)
```

Use `rng` (not `np.random.seed`) for all random sampling in this assignment.

Draw order matters. A fresh `default_rng(99)` produces one long stream of numbers, handed out in the order you ask for them. Draw the inputs in the order the table below lists them, every time. Drawing them in a different order, or drawing only one of them, gives different values — and a different answer.

Track A — CE: Annual Highway Maintenance Cost

A transportation engineer manages **80 km** of rural highway. Each year, the total maintenance cost depends on three uncertain quantities:

#	Input	Distribution	Interpretation
1	potholes_per_km	Normal(3.5, 0.9)	Road condition varies by winter severity
2	cost_per_pothole	Normal(380, 60) \$/pothole	Labor and material cost uncertainty
3	overhead_rate	Normal(0.14, 0.025)	Contractor overhead markup

Annual cost model:

$\text{annual_cost} = 80 \times \text{potholes_per_km} \times \text{cost_per_pothole} \times (1 + \text{overhead_rate})$

The district's annual maintenance budget is **\$115,000**.

Track B — ArcE: Demand Response Savings

A building owner is evaluating a **demand response (DR) program**: during summer peak hours, they will reduce cooling by curtailing air supply. The annual savings depend on four uncertain factors:

#	Input	Distribution	Interpretation
1	peak_hours	Normal(620, 50) hr/yr	Hours when utility charges peak rate
2	curtailment_eff	Normal(0.18, 0.04) fraction	% cooling load shed during DR events
3	peak_rate	Normal(0.22, 0.03) \$/kWh	Utility peak demand rate
4	off_peak_rate	Normal(0.10, 0.01) \$/kWh	Rate during non-DR hours

From the Week 10 regression, **average cooling during BH hours ≈ 74.1 kWh/hr.**

Annual DR savings model:

$$\text{savings} = \text{peak_hours} \times (74.1 \times \text{curtailment_eff}) \times (\text{peak_rate} - \text{off_peak_rate})$$

This is the cost avoided by shifting load from the peak rate to the off-peak rate. The DR program has a fixed participation cost of **\$800/year**.

Everything below applies to **your track only**. Where the text says “the output”, that means `annual_cost` for Track A and `savings` for Track B; “the threshold” means the \$115,000 budget for Track A and the \$800 participation cost for Track B.

Problem 1 — Simulating the Distribution (30 pts)

P1a (8 pts) — Centre and spread

Draw all inputs for $N = 10,000$ scenarios, in table order, and compute the output for each.

ANSWER_P1a_mean = ____

ANSWER_P1a_std = ____

In one sentence below your code: the deterministic estimate — every input at its mean, no simulation — gives a single number. Is your simulated mean close to it, and would you expect it to be?

P1b (8 pts) — The interval and the threshold

ANSWER_P1b_p10 = ____ # 10th percentile
ANSWER_P1b_p90 = ____ # 90th percentile
ANSWER_P1b_prob = ____ # $P(\text{output exceeds the threshold})$, 4 decimals

P10 and P90 bracket the middle 80% of outcomes.

P1c (7 pts) — How many runs is enough?

The lecture asserted that $N = 10,000$ is not arbitrary. Test it. **One run of $N = 100$ does not show this** — a single small sample can land anywhere, and yours might land close to the truth by luck. What matters is how far the answer moves *from run to run*.

For each sample size $N \in \{100, 1,000, 10,000\}$, repeat the whole simulation **20 times** using seeds 1000, 1001, ... 1019 — a fresh `np.random.default_rng(seed)` each time — and record the P90 of the output. Then report the **spread** (largest minus smallest) of those 20 P90 estimates:

ANSWER_P1c_spread_100 = ____ # max - min of the 20 P90 estimates at $N=100$
ANSWER_P1c_spread_1000 = ____
ANSWER_P1c_spread_10000 = ____

Round each to the nearest whole unit. In one sentence: roughly what happens to the spread each time N goes up by a factor of 10, and does that match the $\sqrt{N}$ rule from lecture?

P1d (7 pts) — Which input drives the uncertainty?

Run a one-at-a-time variance decomposition. For each input in turn: draw **all** inputs in table order from a fresh `np.random.default_rng(99)`, then replace every *other* input with its distribution mean, and take the variance of the resulting output. That input's share is its variance divided by the variance of your full P1a simulation, as a percentage.

ANSWER_P1d_dominant_input = ____ # the input name, as a string
ANSWER_P1d_dominant_pct = ____ # its share of total variance, 1 decimal
ANSWER_P1d_smallest_pct = ____ # the smallest input's share, 1 decimal

The shares will not sum to exactly 100%. In one sentence, say why not.

Problem 2 — The Design Decision (18 pts)

P2a (6 pts) — Sizing to a risk level, not to an average

Your client does not want a budget that is exceeded half the time. Report the value that leaves only a **5% chance** of being exceeded:

ANSWER_P2a_p95 = ____

For Track A this is the budget the district would need to be 95% confident of staying within. For Track B it is the savings figure the owner could quote as a near-best case.

P2b (6 pts — written)

In 3–5 sentences, explain the P10–P90 range from P1b to your client in plain language — a district engineer or a building owner, not a statistician. State both numbers. Then answer the question they will actually ask: **is this a wide range or a narrow one?** Justify with a ratio or a percentage of the mean, not an adjective.

P2c (6 pts — written)

Your P1d decomposition says one input dominates. Suppose you have a budget to reduce the uncertainty in exactly **one** input — better road surveys for Track A, a metered pilot study for Track B.

In 3–5 sentences: which input do you buy information about, and why? Cite your dominant and smallest percentages. Then address the trap: an input can have a large *uncertainty* and still barely matter, or a small uncertainty and matter a lot. Explain which of those is happening to the input you are **not** buying.

Problem 3 — Reflection (12 pts)

100–150 words, typed prose, not bullet points. Cite at least two specific numbers from your own simulation.

Week 12 ranked the inputs with a tornado diagram — a one-at-a-time method. Week 13 ran a full Monte Carlo, varying everything at once. Both told you which input matters most, and this week they agreed.

Describe one engineering situation where OAT would have been enough, and one where it would have given the wrong answer and only Monte Carlo would have caught it. Be concrete about *what feature of the problem* makes the difference — not “MC is more accurate,” but what specifically OAT cannot see.

Grading Summary

Problem	Description	Points
P1a	Mean and standard deviation of the simulated output	8
P1b	P10, P90, and exceedance probability at the threshold	8
P1c	Convergence: spread of the P90 estimate at three sample sizes	7
P1d	OAT variance decomposition — dominant input and its share	7
P2a	The 5%-risk design value (P95)	6
P2b	Written: the P10–P90 range explained to the client	6
P2c	Written: which input to buy information about, and why	6
P3	Reflection: when OAT suffices and when it misleads	12
Total		60

Deliverable

Submit a single Colab notebook (.ipynb) with all cells executed. Every ANSWER_ variable must appear in cell output. Written responses go in markdown cells immediately after the code cell they discuss.